Appendix: COMPANY Database Schema and Attribute Descriptions

Reference for HW#2: Relational Database and Relational Algebra

Purpose. This appendix explains the meaning of each attribute in the COMPANY database schema used in the homework. Students should use these descriptions to understand the data before writing Relational Algebra expressions.

Notation. PK = Primary Key, FK = Foreign Key. A composite primary key contains more than one attribute.

Schema Overview

Employee(Fname, Lname, Ssn, Address, Salary, Dno, SuperSsn)

Department(Dname, Dnumber, MgrSsn, MgrStartDate)

Project(Pname, Pnumber, Plocation, Dnum)

Works_on(Essn, Pno, Hours)

Dependent(Essn, DependentName, Sex, Bdate, Relationship)

1. Employee

Stores information about each employee in the company.

Primary Key: Ssn

Attribute	Suggested Type	Key / Reference	Description
Fname	VARCHAR(50)	-	Employee's first name.
Lname	VARCHAR(50)	-	Employee's last name.
Ssn	CHAR/VARCHAR	PK	Unique identification number for an employee.
Address	VARCHAR(255)	-	Employee's home or contact address. It may include city information, such as Bangkok.
Salary	DECIMAL	-	Employee's salary amount.
Dno	INT	FK ⟶ Department.Dnumber	Number of the department in which the employee works.
SuperSsn	CHAR/VARCHAR	FK ⟶ Employee.Ssn	SSN of the employee's direct supervisor. This is a self-referencing foreign key and may be NULL for an employee with no supervisor.

2. Department

Stores information about the departments in the company.

Primary Key: Dnumber

Attribute	Suggested Type	Key / Reference	Description
Dname	VARCHAR(100)	Candidate/alternate unique value*	Name of the department, for example Research or Administration.
Dnumber	INT	PK	Unique number assigned to a department.
MgrSsn	CHAR/VARCHAR	FK ⟶ Employee.Ssn	SSN of the employee who manages the department.
MgrStartDate	DATE	-	Date on which the current manager started managing the department.

*** Note:** In many versions of this schema, department names are also assumed to be unique. For this homework, use Dnumber as the primary key.

3. Project

Stores information about projects carried out by the company.

Primary Key: Pnumber

Attribute	Suggested Type	Key / Reference	Description
Pname	VARCHAR(100)	Candidate/alternate unique value*	Name of the project.
Pnumber	INT	PK	Unique number assigned to a project.
Plocation	VARCHAR(100)	-	Location where the project is carried out, such as Bangkok or Chiang Mai.
Dnum	INT	FK ⟶ Department.Dnumber	Number of the department that controls or is responsible for the project.

*** Note:** In many versions of this schema, project names are also assumed to be unique. For this homework, use Pnumber as the primary key.

4. Works_on

Represents the many-to-many relationship between employees and projects. Each row means that one employee works on one project.

Composite Primary Key: (Essn, Pno)

Attribute	Suggested Type	Key / Reference	Description
Essn	CHAR/VARCHAR	PK, FK ⟶ Employee.Ssn	SSN of the employee who works on the project.
Pno	INT	PK, FK ⟶ Project.Pnumber	Project number of the project on which the employee works.
Hours	DECIMAL	-	Number of hours per week that the employee works on that project.

5. Dependent

Stores information about dependents associated with employees.

Composite Primary Key: (Essn, DependentName)

Attribute	Suggested Type	Key / Reference	Description
Essn	CHAR/VARCHAR	PK, FK ⟶ Employee.Ssn	SSN of the employee to whom the dependent belongs.
DependentName	VARCHAR(100)	PK	Name of the dependent. It is unique only within the dependents of a particular employee under this schema assumption.
Sex	CHAR(1)	-	Sex of the dependent, for example M or F.
Bdate	DATE	-	Date of birth of the dependent.
Relationship	VARCHAR(30)	-	Relationship of the dependent to the employee, such as Spouse, Son, or Daughter.